

Roll No.

DEDE-01

**B. TECH. (MECHANICAL/AUTOMOBILE ENGG./ROBOTICS)
(SIXTH SEMESTER)**

MID SEMESTER EXAMINATION, April, 2023

DESIGN FOR SUSTAINABLE DEVELOPMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the sustainability aspects of 3D printing, considering its advantages and disadvantages. (CO1, CO2)

OR

- (b) Mechanical engineering will help you in achieving goals defined by United Nations. Explain in detail with *five* goals related to your branch.

(CO1, CO2)

2. (a) Explain with the help of diagram the sustainability circle of your city.

(CO1, CO3)

OR

- (b) Explain cradle to cradle and biomimicry design approaches with one example each. (CO1, CO3)

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(2)

3. (a) Explain spatio-social innovation level with *two* examples. (CO1, CO2)

OR

- (b) Explain the design for the base of pyramid with all four points.

(CO1, CO2)

4. (a) What are the challenges of product service system design ? Explain with *ten* points. (CO2, CO3)

OR

- (b) Explain Product life cycle assessment in detail with *two* examples.

(CO2, CO3)

5. (a) You wish to design one sustainable product for your city. Name the product and explain in detail. (CO4, CO2)

OR

- (b) Explain open loop recycling in detail with the help of a diagram.

(CO4, CO2)

Roll No.

TMS-601

B. TECH. (ME-RA) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
AUTOMOTIVE MECHATRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by Mechatronics ? Give examples of various mechatronic systems used in industries, automobiles, aircrafts and domestic purpose. (CO1)

OR

- (b) What are the essential components of a mechatronic system ? Briefly explain them with a block diagram. (CO1)

2. (a) Give different classifications of automobile based on use, fuel, wheels, drive axles and transmission. (CO1)

OR

- (b) Briefly explain about the following automobile aggregates : (CO1)

- (i) Power train
(ii) Comfort system.

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3. (a) Explain about the typical sensor measurement system with a block diagram. (CO1 & CO2)

OR

- (b) What are the important operational factors in sensor selection ? (CO1 & CO2)

4. (a) Explain about the working and use of a Hall effect sensor with a neat diagram. (CO2)

OR

- (b) Discuss about the following types of sensors : (CO2)

- (i) Active
- (ii) Passive
- (iii) Analog
- (iv) Digital

5. (a) Explain about the working and purpose of an oxygen sensor in cars with a neat diagram. (CO2)

OR

- (b) Explain about the working and purpose of a mass air flow sensor in cars with a neat diagram. (CO2)

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TAS-601

B. TECH. (ME) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
AIRCRAFT PROPULSION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the principle of jet propulsion. (CO2)

OR

- (b) What type of compressor is there in the gas turbine engine ? Explain in detail. (CO2)

2. (a) Draw a typical temperature, pressure plot of a gas turbine engine. (CO2)

OR

- (b) What is the function of stator blades in the compressor ? (CO2)

3. (a) Briefly explain the difference between Turbojet engine and Turbo Prop Engine. (CO1)

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OR

(b) Why is a turbofan better than a turbojet ? (CO1)

4. (a) What is the use of combustion chamber in gas turbine ? (CO1)

OR

(b) What are the different types of combustion chamber in jet engine ? (CO1)

5. (a) Explain in detail the thermodynamic cycle of a typical gas turbine engine. Draw P-v and T-s diagrams. (CO2)

OR

(b) What is the function of low pressure compressor in gas turbine engine ? (CO1)

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DEDE-06

B. TECH. (ME) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
MODELLIGN OF CONTROL SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.
(iii) Draw neat diagrams as necessary.
(iv) Make assumptions for any missing data.

1. (a) Explain the basic elements of a control system. (CO1)

OR

- (b) Explain the following with examples from engineering domain : (CO1)

- (i) Linear and nonlinear control systems
(ii) Time variant and time invariant control systems

2. (a) Compare open loop and closed loop control systems. (CO1)

OR

- (b) What is the need for modelling of a control system ? Explain with relevant example. (CO1)

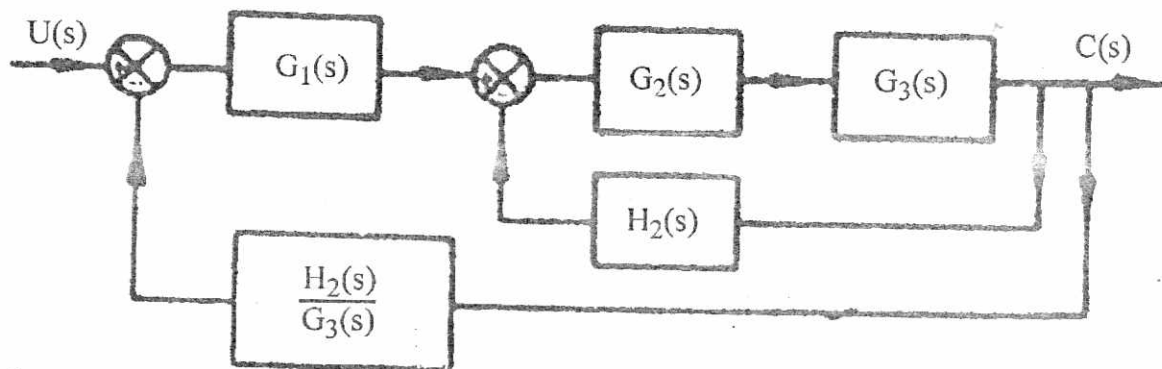
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3. (a) Derive the differential equation of a mechanical system consisting of a spring mass and damper and draw the response to a step input. (CO2)

OR

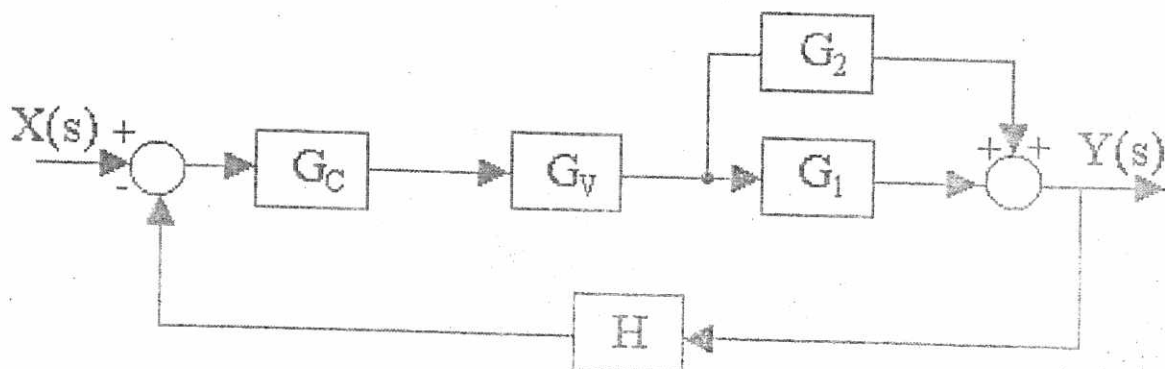
- (b) Derive the differential equation of an electrical system consisting of a resistance, inductance and capacitance element and draw the response to a step input. (CO2)

4. (a) Derive the Equivalent open loop transfer function : (CO2)



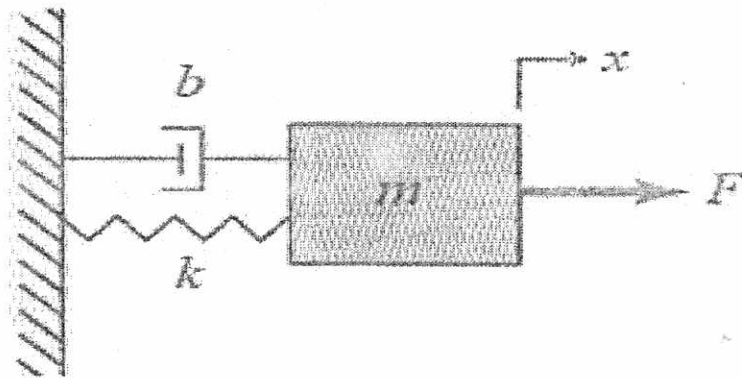
OR

- (b) Derive the Reduced form of the following block diagram : (CO2)



(3)

5. (a) Obtain the solution of the following system using Laplace transforms : (CO2)



For $F = 1$ N, $m = 1$ kg, $b = 9$ N/m/s, $k = 20$ N/m, and all zero initial conditions.

OR

- (b) Derive the transfer function of a simple RLC circuit with zero initial conditions. (CO2)

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DEEM-05

B. TECH. (ME/AME/ROBOTICS) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

OPERATIONS RESEARCH

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain how and why operations research methods have been valuable in aiding executive decisions. (CO1)

OR

- (b) What is meant by a mathematical model of a real situation ? Discuss the importance of models in the solution of OR problems. (CO1)

2. (a) A company sells two different products A and B, making a profit of ₹ 40 and ₹ 30 per unit, respectively. They are both produced with the help of a common production process and are sold in two different markets. The production process has a total capacity of 3000 man-hours. It takes three hours to produce a unit of A and one hour to produce a unit of B. The market has been surveyed and company officials feel that the maximum number of units of A that can be sold is 8000 units and

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that of B is 12000 units. Subject to these limitations, products can be sold in any combination. Formulate this problem as an LP model to maximize profit and solve it graphically. (CO2)

OR

- (b) A firm makes two products X and Y and has a total production capacity of 9 tonnes per day. Both X and Y require the same production capacity. The firm has a permanent contract to supply at least 2 tonnes of X and at least 3 tonnes of Y per day to another company. Each tonne of X requires 20 machine hours of production time, and each tonne of Y requires 50 machine hours of production time. The daily maximum possible number of machine hours is 360. All of the firm's output can be sold. The profit made is ₹ 80 per tonne of X and ₹ 120 per tonne of Y. Formulate this problem as an LP model and solve it by using graphical method to determine the production schedule that yields the maximum profit. (CO2)

3. (a) Solve the following LP problem using the Simplex Method : (CO2)
Minimize :

$$Z = x_1 - 3x_2 + 2x_3$$

Subject to the constraints :

$$3x_1 - x_2 + 3x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0.$$

OR

(b) Use penalty (Big-M) method to solve the following LP problem :

(CO2)

Minimize :

$$Z = 5x_1 + 2x_2 + 10x_3$$

Subject to :

$$x_1 - x_3 \leq 10$$

$$x_2 + x_3 \geq 10$$

$$x_1, x_2, x_3 \geq 0.$$

4. (a) A steel company has three open hearth furnace and five rolling mills. The transportation costs (rupees per quintal) for shipping steel from furnaces to rolling mills are given in the following table : (CO3)

		Rolling Mills				
		M1	M2	M3	M4	M5
Furnance	F1	4	2	3	2	6
	F2	5	4	5	2	1
	F3	6	5	4	7	7
Demand		4	4	6	8	8

Use VAM to find the Transportation Cost.

OR

- (b) A firm manufacturing a single product has three plants I, II and III. They have produced 60, 35 and 40 units, respectively during this month. The firm had made a commitment to sell 22 units to customer A, 45 units to customer B, 20 units to customer C, 18 units to customer D and

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30 units to customer E. Find the minimum possible transportation cost of shifting the manufactured product to the five customers. The net unit cost of transporting from the three plants to the five customers is given below :

(CO3)

		A	B	C	D	E
Plants	I	4	1	3	4	4
	II	2	3	2	2	3
	III	3	5	2	4	4

5. (a) A pharmaceutical company producing a single product sold it through five agencies situated in different cities. All of a sudden, there rouse a demand for the product in another five cities that didn't any agency of the company. The company is now facing the problem of deciding on how to assign the existing agencies in order to despatch the product to needy cities in such a way that the travelling distance is minimized. The distance between the surplus and deficit cities (in km) is given in the following table.

(CO3)

		Deficit Cities				
		a	b	c	d	e
Surplus Cities	A	160	130	115	190	200
	B	135	120	130	160	175
	C	140	110	125	170	185
	D	50	50	80	80	110
	E	55	35	80	80	105

Determine the optimum assignment schedule.

OR

- (b) A company has four machines that are to be used for three jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given in the following table : (CO3)

		Machines			
		W	X	Y	Z
Jobs	A	18	24	28	32
	B	8	13	17	18
	C	10	15	19	22

What are the job assignment pairs that shall minimize the cost ?

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DEEM-03

B. TECH. (ME/AME/ROBOTICS) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MANAGEMENT OF INTELLECTUAL PROPERTY RIGHTS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by intellectual property and intellectual property rights ? (CO1 & CO3)

OR

- (b) Classify various types of intellectual property rights. Does Indian Patenting system give worldwide protection ? (CO1 & CO3)

2. (a) What types of inventions are not patentable in India ? (CO1)

OR

- (b) What can be patented ? Write down the criteria of patentability. (CO1)

3. (a) Give an overview of intellectual property management. Why is it important for innovation driven organizations ? (CO2 & CO4)

OR

- (b) What are the forms required to be filled for filing a patent application in India ? (CO2 & CO4)

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4. (a) Discuss the important principle provisions of the Berne Convention for the protection of Literary and Artistic works. (CO2)

OR

- (b) Is there any provision in the Patents Act for early publication ? Explain If it is there. (CO2)

5. (a) Explain the situations where valuation of IP is important for an organization. (CO2 & CC3)

OR

- (b) Write down the stages of intellectual property asset creation and explain any *one* stage in brief. (CO2 & CO3)

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TME-609

B. TECH. (ME) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

INDUSTRIAL ENGINEERING AND PROJECT MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) From the following data, you are required to calculate break-even point and net sales value at this point : (CO1)

	₹
Direct material cost per unit	10
Direct labour cost per unit	5
Fixed overhead	50,000
Variable overheads @ 60% on direct labour :	
Selling price per unit	25
Trade discount	4%

If sales are 10% and 25% above the break even volume, determine the net profits.

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OR

- (b) From the following particulars, find out the break-even-point : (CO1)

	₹
Variable Cost per unit	15
Fixed Expenses	54,000
Selling Price per unit	20

What should be the selling price per unit, if the break-even point should be brought down to 6000 units ?

2. (a) A small project consisting of eight activities has the following characteristics : (CO1)

Activity	Preceding activity	Most optimistic time (a)	Most likely time (m)	Most Pessimistic time (b)
A	None	2	4	12
B	None	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	B, C	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

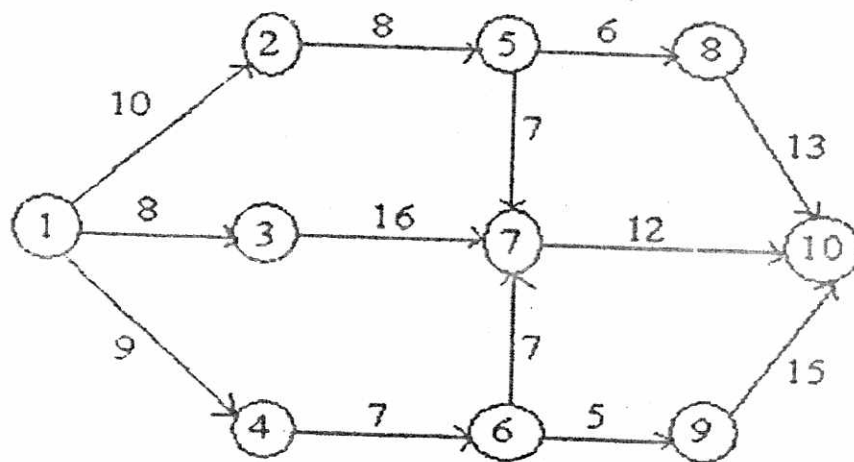
- (i) Draw the PERT network for the project.
(ii) Prepare the activity schedule for the project.

(3)

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OR

- (b) Explain the CPM in network analysis. What are the rules for drawing network diagram ? Also mention the common errors that occur in drawing networks. (CO1)
3. (a) Determine the early start and late start in respect of all node points and identify critical path for the following network : (CO1)



OR

- (b) Explain the following terms : (CO1)
- (i) Optimistic time
 - (ii) Most likely time
 - (iii) Pessimistic time
 - (iv) Expected time
 - (v) Variance
4. (a) What is ABC analysis ? Write the steps involved in ABC analysis of inventory management. (CO2)

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OR

- (b) What do you mean by Productivity ? Explain in detail the categories of production. (CO2)
5. (a) What is PPC ? Write the importance of Production Planning and Control in detail. (CO2)

OR

- (b) Discuss the terms forecasting, routing, scheduling, dispatching and follow up in details with suitable examples. (CO2)

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TME-606

B. TECH. (ME) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
COMPUTER AIDED MANUFACTURING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss salient features CAM. Describe NC Machine tool. (CO1)

OR

(b) Explain working principle of CNC machine tools with neat schematic diagram. Classify basic elements of CNC. (CO1)

2. (a) Compare CAM against conventional Manufacturing Processes. Discuss their advantages and limitations. (CO1)

OR

(b) Appraise advantages of NC and CNC machine tools over conventional machine tools. (CO1)

3. (a) Explain working principle of DNC systems with a neat sketch. (CO2)

OR

(b) Distinguish between Dire Control and Distributed Numerical Control with a neat sketch and discuss their salient features. (CO2)

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4. (a) Enunciate advantages of DNC compared to NC and CNC. (CO2)

OR

- (b) Explain communication principle employed in DNC systems. (CO2)

5. (a) Interpret and demonstrate Group Technology (GT) features. (CO3)

OR

- (b) Classify part families employed in Group Technology. Explain part classification systems with brief details. (CO3)

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TME-602

B. TECH. (MECHANICAL ENGG.) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

DESIGN OF MACHINE ELEMENTS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Explain the phenomenon of surging in open coiled helical spring.
(ii) Define spring index, spring stiffness, solid length, free length, pitch of open coiled helical spring. (CO1)

OR

- (b) What are the functions of concentric open coiled helical helical spring ?
Derive the expression of radial clearance in terms of spring index if it is kept as the difference of wire radius. (CO1)
2. (a) A railway wagon moving at a velocity of 1.5 m/s is brought to rest by a bumper consisting of two helical springs arranged in parallel. The mass of the wagon is 1500 kg. The springs are compressed by 150 mm in bringing the wagon to rest. The spring index can be taken as 6. The springs are made of oil-hardened and tempered steel wire of SW grade

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($G = 81370 \text{ N/mm}^2$) . The constants A and m can be taken as 1855 and 0.187 respectively. The permissible shear stress for the spring wire can be taken as 50% of the ultimate tensile strength. Design the spring and calculate : (i) wire diameter; (ii) mean coil diameter; (iii) number of active coils; (iv) total number of coils; (v) solid length; (vi) free length

(CO1)

OR

- (b) A semi-elliptical laminated spring is to carry a load of 5000 N and consists 8 leaves 46 mm wide, two of the leaves being of full length. The spring is to be made 1000 mm between the eyes and is held at the centre by a 60 mm wide band. Assume that the spring is initially stressed so as to induce an equal stress of 500 N/mm^2 when fully loaded. Design the spring giving (i) thickness of leaves (ii) eye diameter (iii) length of leaves (iv) Maximum deflection and camber.

Assume $E = 2.1 \times 10^5 \text{ N/mm}^2$. (CO1)

3. (a) (i) What are the important factors considered for the selection of flat belt for power transmission ?
(ii) Explain the advantages and disadvantages of V-belt drive over Flat belt drive. (CO2)

OR

- (b) Explain the limitation to maximum and minimum number of teeth on sprocket in a chain drive. (CO2)
4. (a) It is required to design a chain drive to connect a 12 kW, 1400 r.p.m. electric motor to a centrifugal pump running at 700 r.p.m. The service conditions involve moderate shocks. (CO2)

(3)

- (i) Select a proper roller chain and give a list of its dimensions.
- (ii) Determine the pitch circle diameters of driving and driven sprockets.
- (iii) Determine the number of chain links.

Pitch			
	Simplex	Duplex	Triplex
9.525	4.1	7.7	10.9
12.7	7.1	13.6	20.2
15.785	9.1	18.2	27.3
19.05	11.7	23.6	35.4

OR

- (b) Design a flat belt to drive a winch from an electric motor of 11 kW power. Speed of motor shaft is 750 rev/min. Speed ratio is 4. Belt position is horizontal and there is considerable variation of load. (CO2)
5. (a) What are the different types of loads considered to design a wire rope for hoisting purposes ? Discuss the design of wire rope with mathematical expressions. (CO2)

OR

- (b) A V-belt drive is to transmit 14.7 kW to a compressor. The motor speed is 1150 rev/min and the compressor pulley runs 400 rev/min. Determine the size and number of belts required. (CO2)